

PART 2: Dialogue Continuity Classification

Mission Background

In developing voice assistants (e.g., Siri or Alexa), the system must determine if a user has finished speaking (**Complete**) or if the utterance is unfinished/interrupted (**Incomplete**).

Robust endpoint detection **facilitates seamless, human-like interaction** by precisely identifying when a user has finished speaking, thereby eliminating awkward silences and preventing premature processing of incomplete thoughts.

This mechanism ensures conversational flow and barge-in capabilities while optimizing computational efficiency by triggering resource-intensive LLM inference only when a complete, actionable request is finalized.

This assignment uses a dataset containing transcriptions of real-world dialogues.

- **Label 1 (Complete):** Semantic intent is finished; the system can respond.
- **Label 0 (Incomplete):** Semantic intent is unfinished; the system should continue listening.

id	content	label
1166	i think we should consider the actually the client prefers the original design	1
1127	Can you reset my password? My account password immediately	1
1240	im wondering if we need to no wait the test cases all passed successfully	1
1853	This code needs to be reviewed by tomorrow, don't you thi	0
194	This homework is taking forever, btw the new superhero movie looks	0

We provide two .CSV file:

- **train.csv** -> model training
- **test.csv** -> kaggle competition

train.csv for training your model. **test.csv** for generating your solution.csv for Kaggle competition. Please use **the train.csv and test.csv file to complete part 2.**

1. Data Balancing (15% Total)

The dataset is naturally imbalanced. You must demonstrate how different sampling techniques affect the model's ability to recognize the minority class.

Code Implementation (8%)

Please complete your code using **part2_code_template.ipynb**.

Since the dataset is imbalanced, models may bias toward the majority class. You must implement and compare two methods. For this part, please choose **the same training model** (TF-IDF + SVM/ XGBoost/ Bert....) for comparing the two data balancing methods :

- **Basic Method (Required):** Random Over-sampling. Duplicate minority class samples until the ratio is 1:1.
- **Advanced Method (Choose 1+):**
 - **EDA (Easy Data Augmentation):** Random synonym replacement, insertion, swapping, or deletion.
 - **Back-translation:** Translate text to another language (e.g., German) and back to English.
 - **SMOTE:** Synthesize minority samples in the vector space.
 - **Cost-Sensitive Learning:** Adjust `class_weight='balanced'` in the model loss function.
 - **Heuristic Splitting:** Split complete sentences into incomplete sentences.
 - Any other method you would like to use.

Written Report (7%)

You must **answer all the questions/requirements** listed below.

For this part, please choose **the same training model** (TF-IDF + SVM/ XGBoost/ Bert....) for comparing the two data balancing methods.

- **Code Screenshots:** Include clear code blocks for both implementations with screenshots in your report. And, **explain your code**.
- **Observations:** Please document your findings, and **explain everything in details**, including
 - The training model you choose to compare the results.
 - Briefly explain about random over-sampling.
 - Explain Macro-F1 first, give a comparison of the Macro-F1 scores resulting from those methods, and why this score.
 - A determination of which method performed better, explain why.
 - An explanation of the logic behind your chosen "Advanced Method."
 - The unique mechanism it uses to help the model learn compared to the random method.
 - **Best Method Identification:** Clearly state your best methods, and the specific parameters applied, explain why it outperformed and what you have tried.
 - Any others you find interesting/ like to discuss.

2. Modeling & Enhancements (25% Total)

Please complete your code using [part2_code_template.ipynb](#).

You are required to move from a traditional machine learning baseline to a more sophisticated architecture.

Code Implementation (10%)

1. **Baseline Model:** Establish a stable baseline using **TF-IDF + SVM (Support Vector Machine)**.
 - **Metric:** Macro-F1 Score (Do not use Accuracy due to class imbalance).
2. **Advanced Method:** Implement more advanced methods to improve your model accuracy. You may explore the following aspects:
 - **Rigor:** Use Stratified K-Fold for validation.
 - **Standardization:** Experiment with Lemmatization or N-grams.
 - **Advanced Models:** Try XGBoost or fine-tune BERT/RoBERTa or other models you like for a significant score boost.
 - **Note that, you don't have to try all of them, but you must at least implement one of the advanced models.**

Written Report (15%)

You must **answer all the questions/requirements** listed below.

Note that you must at least implement one of the advanced models.

- **Code Screenshots:** Include clear code blocks for both implementations with screenshots in your report. And, **explain your code**.
- **Observations:** Please document your findings, and **explain everything in details**, including:
 - A comparison of the Macro-F1 scores resulting from those methods, and explain why this score.
 - Briefly explain what TF-IDF and SVM is.
 - A determination of which method performed better, explain why..
 - An explanation of the logic behind your chosen "Advanced Method."
 - The unique mechanism it uses to help the model learn.
 - **Best Model Identification:** Clearly state your best model, the methods used, and the specific parameters applied., explain why it outperformed and what you have tried.
 - Any others you find interesting/ like to discuss.

3. Kaggle Benchmarking (10% Total)

Your final model must be tested against the private leaderboard to validate its real-world performance.

Benchmarking (6%)

- **Requirement:** Your submission must achieve a **Macro-F1 score** that exceeds the established **TA Baseline**.

Top Tier Bonus (4%)

- **Criteria:** Additional marks (up to 4%) will be awarded based on your relative standing on the leaderboard.

4. Peer Evaluation

For the peer evaluation table, each group may rate member contributions on a 1-10 scale. You may decide the score based on your own group collaboration, but the table should briefly indicate each member's contribution.

5. Submission Format Requirements

To ensure your assignment is graded, please follow these formatting rules:

File Format:

- Submit your code and report to E3 separately.
- Please note that each group should **upload only one set** of files in total.
- **Report:** Submit in **.pdf** format.
 - **Filename:** `CLASSNUMBER_GROUPNUMBER_HW3_report.pdf`
 - For examples: `515512_Group01_HW3_report.pdf`
 - Please refer to E3 to check if you're from class **515512** or **515513**.
 - **Requirement:** **Include all members' student IDs and names at the very beginning of the report.**
 - You should **only submit ONE report** for the entire homework.
- **Code:** For Part 2 code, submit in **.ipynb** format.
 - **Filename:** `part2.ipynb`
 - For examples: `part2.ipynb`
- **Code_Entire:** For entire code, submit in **.zip** format.
 - **Filename:** `CLASSNUMBER_GROUPNUMBER_HW3_code.zip`
 - For examples: `515512_Group01_HW3_code.zip`

Please follow the submission layout:

None

```
{515512|515513}_{groupNum}_HW3_code.zip
└─ part1/
   └─ task1/
      └─ task1.ipynb
   └─ task2/
      └─ inference.py
      └─ train_command.txt
      └─ requirements.txt
```

```
└─ (additional Task 2 files)
└─ part2/
    └─ part2.ipynb

{515512|515513}_{groupNum}_HW3_report.pdf
```

- Incorrect file formats will result in **a score of 0** for this homework.
- Doesn't include all members' student ids and names at the very beginning for both code and report will result in **a score of 0** for this homework.

Kaggle Submission Format:

- Submit your result to Kaggle.
- Please note that each group should **only have one group name** in total.
- **Results:** Submit in **.csv** format.
- **Group Name:** **CLASSNUMBER_GROUPNUMBER**
 - For example: **515512_01**.
 - Please refer to E3 to check if you're from class **515512** or **515513**.
- Incorrect group name will result in **a score of 0** for this homework.

Submission Deadline:

- Upload to E3 by: **2026/05/19 (Tue) 23:59**
- Upload to Kaggle by: **2026/05/19 (Tue) 23:59**
- **Late submissions for any reason are not allowed. Late submissions are judged by the E3 system. Please upload your homework early.**
- Late submissions will result in **a score of 0** for this homework.

You can find all file you need for part 2 here:

- ❖ train.csv: [train.csv](#)
- ❖ test.csv: [test.csv](#)
- ❖ part2 code template: [part2_code_template.ipynb](#)
- ❖ report template: [HW3_report_template](#)
- ❖ part 2 kaggle: <https://www.kaggle.com/competitions/dialogue-continuity-classification>
- ❖ Kaggle Invitation link (Please use this link to join the competition if you have already provided email to the TAs) :
<https://www.kaggle.com/t/be85b84fe61e5d3e74c10da444623101>
- ❖ part 2 introduction: [part2_introduction](#)